

FICTITIOUS PRODUCT — DEMONSTRATION DATA

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TECHNICAL SPECIFICATION DATASHEET

Industrial Passive 3-Wire Pt100 RTD Temperature Immersion Probe in AISI 316L Stainless Steel

SKU:	CN-DEMO-PT100-3W-A1	Model:	CN-RTD-P1
Category:	Primary Temperature Sensors / Resistance Temperature Detectors (RTD)	Manufacturer / Rev:	ControlNet Demo Systems Rev 1.0 (Demonstration)

Section 1: Identification and Model Overview

The primary temperature sensor model **CN-RTD-P1** (SKU code: **CN-DEMO-PT100-3W-A1**) is a passive 3-wire Pt100 RTD immersion probe with a platinum wire-wound sensing element for immersion in thermal fluids and industrial processes. It strictly conforms to international resistance-versus-temperature characteristics under IEC 60751.

Identification Parameter	Catalog Specification
SKU Code	CN-DEMO-PT100-3W-A1
Model Designation	CN-RTD-P1
Device Classification	Immersion probe / passive 3-wire Pt100 RTD (NO integrated transmitter, NO digital interface)
Operating Principle	Pure platinum resistance thermometry with temperature-dependent electrical resistance variation
Engineering Applications	Direct temperature measurement in process piping, heat exchangers, fluid tanks, and thermowells

Section 2: Physical Mounting and Mechanical Form Factor

Corrosion-resistant mechanical construction engineered for leak-tight immersion in pressurized liquid or gaseous media.

Mechanical Parameter	Nominal Specification and Mounting Attributes
Mounting & Process Connection	Direct threaded process mounting via fixed 1/2" NPT male fitting
Sheath Material	Austenitic stainless steel grade AISI 316L (1.4404) sheath of high chemical purity
Probe Dimensions	Outer diameter: 6.0 mm Usable immersion length in process: 150 mm
Static Pressure Rating	Permissible hydrostatic operating pressure up to 40 bar at 20 °C (25 bar at 200 °C)
Lead Cable & Extension	2.0-meter flexible shielded cable with oil- and heat-resistant silicone/Teflon jacket
Connection Terminations	3 flexible stranded conductors with tinned copper ferrules: 2 red (common return) and 1 white

Section 3: Electrical Power Supply & Operational Tolerances

Excitation specifications for receiving instruments. Purely passive sensing device with no autonomous power consumption.

Electrical Parameter	Passive Sensor Operating Conditions
Internal Power Supply	NO INTERNAL POWER SUPPLY (0 VDC / 0 VAC). Purely passive 3-wire Pt100 component
Nominal Excitation Current	0.1 mA to 1.0 mA DC constant current (supplied externally by measurement bridge or PLC/PID input)
Maximum Safe Current	2.0 mA DC (strict thermal threshold to prevent measurement error caused by Joule self-heating)
Dielectric Insulation Resistance	Insulation resistance > 100 MΩ at 500 VDC between internal conductors and metallic sheath
Permissible Voltage Input	DIRECT VOLTAGE PROHIBITED. Applying direct voltage will instantly destroy the platinum sensor element

Section 4: Analog & Discrete I/O Interfaces and Sensors

Platinum resistance characteristics and metrological calibration tolerances.

Metrological Parameter	RTD Sensor Specification
Sensing Element	Pure platinum wire-wound / thin-film Pt100 element compliant with DIN EN 60751
Base Resistance at 0 °C	R0 = 100.00 Ω nominal at 0.00 °C (temperature coefficient alpha α = 0.003850 Ω/Ω/°C)
Metrological Accuracy Class	Class A according to IEC 60751: Tolerance ±(0.15 + 0.002· t) °C (e.g., ±0.15 °C at 0 °C, ±0.35 °C at 100 °C)
Operating Temperature Range	-50.0 °C to +350.0 °C continuous operating temperature on the AISI 316L sheath
Wiring Configuration	Passive 3-wire Pt100 with dual common lead wires for line-resistance compensation
Integrated Electronic Transmitter	NO INTEGRATED TRANSMITTER (0 transmitters). Does not include signal conditioning electronics
Electrical Output Signal	Pure passive ohmic resistance varying with temperature. No 4–20 mA on its own.

Section 5: Communications, Fieldbus and Protocol Specifications

Pure electromechanical transducer characteristics and digital integration requirements.

Fieldbus Parameter	Compatibility Status & Connectivity Specification
Digital Serial Interface	NO DIGITAL INTERFACE (0 interfaces). The probe contains no microprocessor, UART, IC, or serial port
Modbus RTU / TCP Support	NOT APPLICABLE / NO MODBUS ON ITS OWN. Has no digital bus communication capability
HART / Fieldbus Support	Not supported natively (passive transducer without digital modulation)
Digital Integration Requirement	To read this probe on a Modbus network or PLC, it must be wired directly to a dedicated RTD input module (such as the CN-PID-T1) or connected to an external head-mount/DIN-rail RTD transmitter converting to Modbus / 4–20 mA.

Section 6: Engineering Constraints and Design Contraindications

Essential technical exclusion criteria for engineering and instrumentation designs:

Critical Constraint / Contraindication	Technical Rationale and Design Exclusion Rule
CRITICAL CONTRAINDICATION 1: NO 4–20 mA ON ITS OWN	Model CN-RTD-P1 is a passive resistive probe WITHOUT AN INTEGRATED TRANSMITTER . Under no circumstances does it generate a signal on its own: No 4–20 mA on its own , nor does it output standard voltage. It is strictly prohibited to connect this probe directly to 4–20 mA analog inputs of a PLC (such as the CN-DIN-PLC-A1) without an intermediate RTD temperature transmitter or signal conditioner.
CRITICAL CONTRAINDICATION 2: NO MODBUS ON ITS OWN / NO DIGITAL INTERFACE	This sensor has NO DIGITAL INTERFACE and NO MODBUS ON ITS OWN . It cannot be polled, addressed, or connected to RS-485 twisted pairs. Attempting to connect it directly to a Modbus network will result in total system failure.
DESTRUCTION HAZARD: DIRECT VOLTAGE PROHIBITED	Never apply mains voltage (230 VAC, 115 VAC) or DC power sources (24 VDC, 12 VDC) to the probe leads. Current exceeding a few milliamperes will instantly vaporize the 100 Ω platinum filament, permanently destroying the sensor.
WIRING RULE: OBSERVE 3-WIRE COLOR CODING	Always connect the two identically colored wires (red) to the compensation terminals of the reader to cancel out measurement error caused by cable run length.